Problem Set 3
ECO3121 - Fall 2022

November 8, 2022

Due 5 PM, 11/18/2022

Download *aghousehold.dta* and *village_rainfall.dta* datasets from the blackboard site and load into STATA. The main data we use is the National Fixed Point Survey (NPFS), which is a nationally representative panel dataset (unbalanced) of roughly 5000 households in 88 villages between 1995 and 2002. It is collected by the Ministry of Agriculture and Rural Affairs of China.

On the blackboard, we also uploaded a second dataset regarding precipitation records in each village in 2001. See the variable list below. Write up the answers to 1) - 9) below. In addition, also attach the do file that you used to answer the following questions.

variable name	type	format	label
vl_id	int	%8.0g	village ID, consistent with household data
lat_o	float	%8.0g	latitude of each village
lon_o	float	%8.0g	longitude of each village
provname	str24	%24s	province name in Chinese
cityname	str33	%33s	city name in Chinese
countyname	str33	%33s	county name in Chinese
year	float	%9.0g	year of the record
av_rain	float	%9.0g	average precipitation in 2001, measured by mm
sd_rain	float	%9.0g	standard deviation of precipitation across months in 2001
z_rain	float	%9.0g	zero score of precipitation in 2001

Question 1

First, limit the sample to households **in year 2001** by running code "keep if year==2001" in Stata. you can generate *variable* yield (output per unit of land) via $\frac{hx95_nh125}{hx95_nh112}$, and variable *fertilizer* (fertilizer application per unit of land) through $\frac{nh6_3}{hx95_nh112}$.

1. The variable *h95_nh269* indicates how many days for household members in each household had been working as temporal migrants (measured by days) in 2001.

   Generate a binary variable regarding household migration decision. It takes the value of 1 if *h95_nh269* is greater than zero, otherwise it is 0. Generate the natural

log of yield and fertilizer application intensity (using log(fertilizer + 1) and log(yield + 1) to smooth zero values.).

Export a table that lists the number of observations, the mean, the standard deviation for the above three variables. Briefly describe what we learn from the table about the sample data. (3 points)

Variable	Obs	Mean	Std. Dev.	Min	Max
migration	5,468	0.454645	0.497984	0	1
logyield	4,200	5.757682	0.598285	0	8.386335
logfertilizer	4,200	4.308324	0.963697	0	8.229777

This table indicates that, on average, 50% of households had migration activities in 2001. The means of logyield and logfertilizer are 5.76 and 4.30 respectively.

2. Run a linear probability model of household migration decision in question (1) on the natural log of yield.
Interpret the result and comment on its statistical significance. (4 points)

The coefficient is .0495 in column (1). It means that a 100% (1%) increase in yield is associated with a 5% (0.05%) increase in migration probability. It's statistically significant at 1% level.

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	q (2)	q (3)	q (6)	q (7)	q (8)	q (9)
logyield	0.0495***	0.1332***			0.4723***	0.3444***
	(0.013)	(0.035)			(0.046)	(0.053)
av_rain			0.0004***			
			(0.000)			
predicted $\widehat{logyield}$				0.4723***		
				(0.040)		
logincome						-0.0954***
						(0.019)
logconsumption						0.1514***
						(0.016)
Constant	0.1774**	-0.8617***	5.2743***	-2.2585***	-2.2585***	-1.9254***
	(0.074)	(0.200)	(0.025)	(0.232)	(0.266)	(0.227)
Observations	4,200	4,200	4,003	4,003	4,003	4,003
R-squared	0.004		0.100	0.033		

Standard errors in parentheses, *** p<0.01, ** p<0.05, * p<0.1

3. Run a probit model of household migration decision in question (1) on the natural log of yield. To compute the marginal effects, run command *"margins, dydx(*)"*

after a probit regression in Stata.
Interpret the results if the value of *logyield* increases from 5 to 8, and comment on its statistical significance. (4 points)

The coefficient before margins command is 0.133 in column (2). After computation, the effect equals 0.053. It means that a 100% (1%) increase in yield is associated with a 5.3% (0.053%) increase in migration probability. It's statistically significant at 1% level.

4. List three plausible arguments why the point estimate in question (2) and (3) could be biased, and the corresponding bias directions (**upwards** or **downwards**) relative to the true causal effect of household's agricultural production on household migration decision. (6 points)

   - Omitted variable bias - positive bias. For example, household's income. High income households are more likely to migrate and are also more likely to have a higher yield.
   - Omitted variable bias - negative bias. For example, household's member composition. A single-member household is more likely (free) to migrate is less likely to have a higher yield.
   - Measurement errors in yield - negative bias. A measure error can always cause attenuation bias (the estimation is more closer to zero than the true parameter).
   - Simultaneity bias (reversal causality). Not sure about the direction of bias. It depends.

5. Now your friend suggests that the fertilizer application intensity of each household could be a valid instrumental variable (IV) for your measure of household's yield. List the assumptions that need to hold true for this to be correct and explain why this instrumental variable is true (or not true). (4 points)

   - Relevance. $Cov(fertilizer, yield) \neq 0$. Fertilizers should have direct effects on yields.
   - Exogeneity. $Cov(fertilizer, \mu) = 0$. Fertilizers should not be correlated with household migration decision.
   - Apparently, the Exogeneity assumption does not hold true. Because the application of fertilizer is correlated with μ, for example, household income is included in μ.

6. Now your TA suggests that the rainfall (precipitation) could be a valid instrumental variable (IV) for your measure of household's yield. Try to merge the household's production dataset and precipitation dataset via *vl_id*, the specific village identifier, using stata command *merge* (Many-to-one merge, type *"help merge"* in Stata for assistance).

merge m:1 vl_id using "D:_rainfall.dta"

drop if _merge==2

You decide to use the average rainfall in 2001 (*av_rain*) as the IV for the household's yield. Verify if the assumption of instrument relevance is satisfied using the first stage regression, and export the results. **Write down the first stage regression model** and interpret the result and statistical significance of your result. (5 points)

The first stage regression is:

$$logyield_i = \pi_0 + \pi_1 * av_rain + \nu_i$$

The regression coefficient is 0.00041 in column (3). It means that a 1 mm (100 mm) increase in precipitation will cause 0.041% (4.1%) increase in yield. It's statistically significant at 1% level.

7. Drop the observations where logyield is missing (*drop if logyield==.*)

   Now use Stata/R to estimate the 2nd stage IV point estimate (using linear probability model) as suggested by the TA, and export your result. **Write down the second stage regression model** and interpret the result and statistical significance of your result. (5 points)

   The second stage regression is:

$$migration_i = \beta_0 + \beta_1 * \widehat{logyield}_i + \mu_i$$

   The regression coefficient is 0.4723 in column (4). It means that a 100% (1%) increase in yield will cause a 47% (0.47%) increase in migration probability. It's statistically significant at 1% level.

8. Now your TA tells you that you can use *ivreg* command directly to replicate the results in question (7). Do you find any difference in the IV estimations (β_{IV}) regarding the coefficients and standard errors relative to (7). (3 points)

   In column (4), the magnitude of the coefficients are the same 0.472 for questions (7) and (8) . But the standard errors are different (0.040 vs. 0.046) since the standard error in the second stage regression in (7) is incorrect due to lacking consideration of first-stage regression.

   In reference to your answer to question (2), is the difference between the linear probability model and IV point estimates as you expected or rather not? (2 points)

   The difference between the linear probability model and IV point estimates is expected (or say not expected) because the linear probability model suffers from omitted variable bias, measurement error, and simultaneity bias.

9. Now your professor suggests that the precipitation in each village is likely correlated with other local characteristics that affect household's migration decision, such as the logarithm of the household consumption (log(nh7_40 + 1)) and the logarithm

of the household income (log(nh7_1 + 1)). Propose a way to verify whether these concerns are relevant and export your regression results in the same table as before. Comment on the results and what they imply about the validity of the instrumental variable strategy. (4 points)

We can add controls to our Iv- 2SLS regression and observe the change in IV estimate. If our IV is valid, adding controls suggested by the professor should not change our estimate too much (because the instrumental variable should not be correlated with anything in the error term of the outcome of interest).

As we can see in column (6) of the table, the IV estimate changes significantly (from 0.472 to 0.344) when including controls, and that suggests the IV is correlated with many variables that also affect migration decision. This strongly points to a violation of the exogeneity. Unfortunately, we do not have a valid instrument for household yield.